

Criterion C: System Overview



Criterion C (6 marks) is part of your project blueprint. Criterion B is your high-level project management plan: decomposing the problem and scheduling the work. Criterion C is your low-level technical design: creating the system diagrams, designing your algorithms, and planning your test cases.

Criterion C: Your Project Blueprint

This page covers the *planning* phase of your project. This helps to form the blueprint that you will follow during development.

- **Criterion C (System Overview):** The "how" of technical design.

Criterion C: The Technical Blueprint (6 Marks)

Goal: To provide enough technical detail for a third-party developer to understand your system and recreate your product.

Recommended Word Count: ~150 words

1. The System Model (Diagrams)

You must create diagrams that show the key components of your system, their relationships, and how they interact. This also includes the design of your user interface (UI).

What Diagram Should You Use?

The right diagram depends on your computational context.

- **For Object-Oriented (OOP) Projects (e.g., Java, Python):** Use **UML Class Diagrams** to show your classes, their attributes, methods, and relationships (inheritance, aggregation). Find out how on the [B 3.1.2 Designing Classes](#) page.
- **For Database Projects:** Use an **Entity-Relationship Diagram (ERD)** to show your tables and their relationships (one-to-one, one-to-many, etc.). You should also show your final,

normalised table structures. Find out how on our [A 3.2.2 Entity relationship diagrams](#) page.

- **For Procedural Projects / Algorithms:** Use **Flowcharts** or **Structure Diagrams** to show the logic and flow of control. Find out how on our [B 1.1.4 Flowcharts](#) page.
- **For All Projects:** Include **UI Design Sketches** (wireframes) to show what your product will look like and how the user will interact with it.

2. Algorithms

For the key components of your system, you must provide the algorithms *before* you code them. This demonstrates your algorithmic thinking.

You can present your algorithms using:

- Pseudocode (most common)
- Flowcharts
- Natural language steps

3. The Testing Strategy

This is your *plan* for testing. It must be a "systematic approach" (TSM, p. 71) that directly links back to your **Success Criteria from Criterion A**.

Tool for the Job: Test Plan Table

The best way to present this is a table with your proposed tests. You must plan tests for valid, invalid, and extreme data.

Test ID	Success Criterion it links to	Description of Test	Test Data	Expected Outcome
T-01	SC-04 (Warns on invalid input)	User enters letters into the 'quantity' field.	"abc"	An error message "Invalid input: Please enter a number." is displayed.
T-02	SC-07 (Login works)	User enters a valid username and password.	User: "admin", Pass: "1234"	The user is logged in and taken to the main menu.
T-03	SC-07 (Login fails)	User enters an invalid password.	User: "admin", Pass: "wrong"	An error message "Login failed: Incorrect username or password." is displayed.
T-04	SC-08 (Handles extreme data)	User tries to add an item with a stock level of 99,999,999.	99999999	The item is added to the database successfully.

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